Análisis Matemático

05- Octubre -2026

* Ecuaciones Principales

$$F_a(t) = F_z(t) + F_L(t) = F_c(t) + F_R(t)$$

$$F_z(t) = \frac{P_a(t) - P_p(t)}{z}$$

$$F_L(t) = \frac{1}{L} \int [P_a(t) - P_p(t)]\, dt$$

$$F_c(t) = C \frac{d\, P_p(t)}{dt}$$

$$F_R(t) = \frac{P_p(t)}{R}$$

* Transformada de Laplace

$$F_a(s) = F_z(s) + F_L(s) = F_c(s) + F_R(s)$$

$$F_z(s) = \frac{P_a(s) - P_p(s)}{z}$$

$$F_L(s) = \frac{P_a(s) - P_p(s)}{Lz}$$

$$F_c(s) = C_s \cdot P_p(s)$$

$$F_R(s) = \frac{P_p(s)}{R}$$

* Procedimiento Algebraico

$$F_Z(s) + F_L(s) = F_C(s) + F_R(s)$$

$$\frac{P_a(s) - P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = Cs\,P_p(s) + \frac{P_p(s)}{R}$$

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s)}{Ls} - \frac{P_p(s)}{Ls} = Cs\,P_p(s) + \frac{P_p(s)}{R}$$

$$P_a(s)\left(\frac{1}{Z} + \frac{1}{Ls}\right) = P_p(s)\left(Cs + \frac{1}{R} + \frac{1}{Ls} + \frac{1}{Z}\right)$$

$$P_a(s)\left(\frac{Ls + Z}{LZs}\right) = P_p(s)\left(\frac{CLRZs^2 + L(R+Z)s + RZ}{RLZs}\right)$$

$$\frac{P_p(s)}{P_a(s)} = \frac{\frac{Ls + Z}{LZs}}{\frac{CLRZs^2 + L(Z+R)s + RZ}{RLZs}}$$

* Resultado

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs + RZ}{CLRZs^2 + L(Z+R)s + RZ}$$

* Estabilidad en lazo abierto

$$CLRZs^2 + L(Z+R)s + RZ = 0$$

$$a = CLRZ \quad ; \quad b = L(Z+R) \quad ; \quad C = RZ$$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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* Hipotenso

$z = 0.010$
$L = 0.001$
$R = 1.000$
$C = 3.500$

$\lambda_1 =$
$\lambda_2 =$

∴ el sistema presenta una respuesta estable y subamortiguada a un escalón unitario de entrada.

* Normotenso

$z = 0.010$
$L = 0.001$
$R = 1.000$
$C = 1.500$

$\lambda_1 = -12.0616$
$\lambda_2 = -55.2717$

∴ el sistema presenta una respuesta estable y sobreamortiguada a un escalón unitario de entrada.

* Hipertenso

$z = 0.010$
$L = 0.001$
$R = 1.000$
$C = 1.000$

$\lambda_1 = -11.1267$
$\lambda_2 = -89.8732$

∴ el sistema presenta una respuesta estable y sobreamortiguada a un escalón unitario de entrada.

* Error en estado estacionario

$$e(s) = \lim_{s \to 0} s \, P_a(s) \left(1 - \frac{P_p(s)}{P_a(s)}\right)$$

$$e(s) = \lim_{s \to 0} s \, \frac{1}{s} \left(1 - \frac{LRs + Rz}{CLRzs^2 + L(R+z)s + Rz}\right)$$

$e(s) = 0$
$e(t) = 0 \, V$

* Modelo de ecuaciones integro-diferenciales

$$F_z(t) + F_L(t) = F_c(t) + F_R(t)$$

$$\frac{P_a(t) - P_p(t)}{z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt = C\,\frac{dP_p(t)}{dt} + \frac{P_p(t)}{R}$$

$$\frac{P_a(t)}{z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt = C\,\frac{dP_p(t)}{dt} = \frac{P_p(t)}{R} + \frac{P_p(t)}{z}$$

$$\left(\frac{1}{R} + \frac{1}{z}\right) P_p(t) = \frac{P_a(t)}{z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt - C\,\frac{dP_p(t)}{dt}$$

$$P_p(t) = \left[\frac{P_a(t)}{z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt - C\,\frac{dP_p(t)}{dt}\right] \frac{Rz}{z+R}$$